



Lab: Network Load Balancer (NLB)

Google Cloud Associate Cloud Engineer | GSP007

Overview

Set up a passthrough Network Load Balancer (NLB) on Google Cloud Platform using 3 Compute Engine VMs running Apache. A Layer 4 NLB routes traffic based on IP addresses and port numbers and does not inspect the content of the traffic.

Objectives

- Configure the default region and zone for resources
- Create multiple web server instances
- Configure a load balancing service
- Configure a forwarding rule to distribute traffic

Task 1 — Set Default Region & Zone

Set the default region and zone so all resources are created in the correct location.

```
gcloud config set compute/region [YOUR-REGION]
gcloud config set compute/zone [YOUR-ZONE]
```

```
Cloud Shell
Terminal (qwklabs-gcp-04-2c17cc2e5e1e) x + v
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwklabs-gcp-04-2c17cc2e5e1e.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
student_02_8896023cda0b@cloudshell:~ (qwklabs-gcp-04-2c17cc2e5e1e)$ gcloud auth list
Credentialed Accounts

ACTIVE: *
ACCOUNT: student-02-8896023cda0b@qwklabs.net

To set the active account, run:
  $ gcloud config set account 'ACCOUNT'

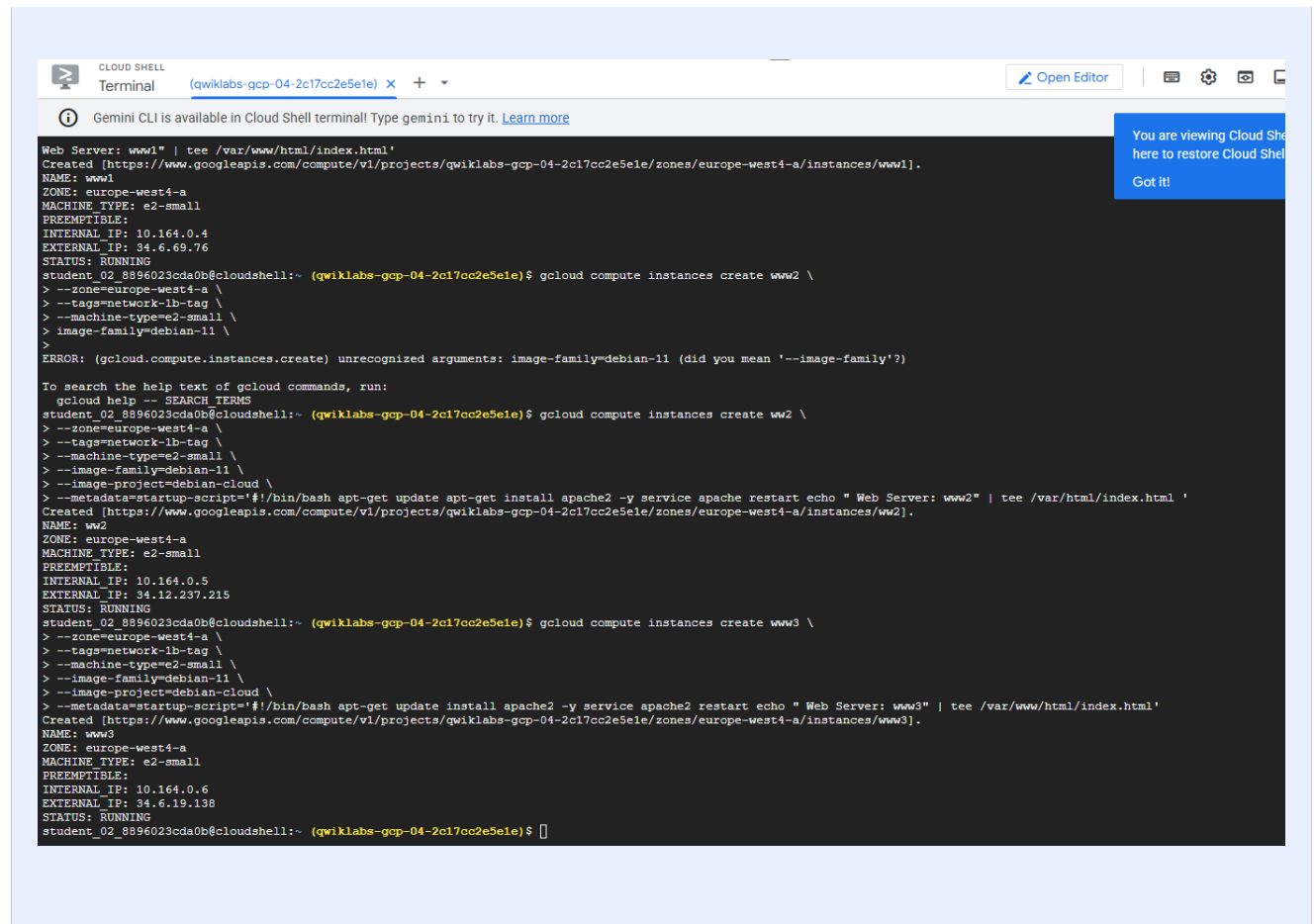
student_02_8896023cda0b@cloudshell:~ (qwklabs-gcp-04-2c17cc2e5e1e)$ gcloud config list project
[core]
project = qwklabs-gcp-04-2c17cc2e5e1e

Your active configuration is: [cloudshell-25370]
[environment: untagged] Read more to tag: g.co/cloud/project-env-tag.
student_02_8896023cda0b@cloudshell:~ (qwklabs-gcp-04-2c17cc2e5e1e)$ gcloud config set compute/region europe-west4
Updated property [compute/region].
student_02_8896023cda0b@cloudshell:~ (qwklabs-gcp-04-2c17cc2e5e1e)$ gcloud config set compute/zone europe-west4-a
Updated property [compute/zone].
student_02_8896023cda0b@cloudshell:~ (qwklabs-gcp-04-2c17cc2e5e1e)$
```

Task 2 — Create Web Server Instances

Created 3 Compute Engine VMs (www1, www2, www3) with Apache installed via startup script. Each VM gets a unique homepage. A firewall rule allows HTTP traffic on port 80.

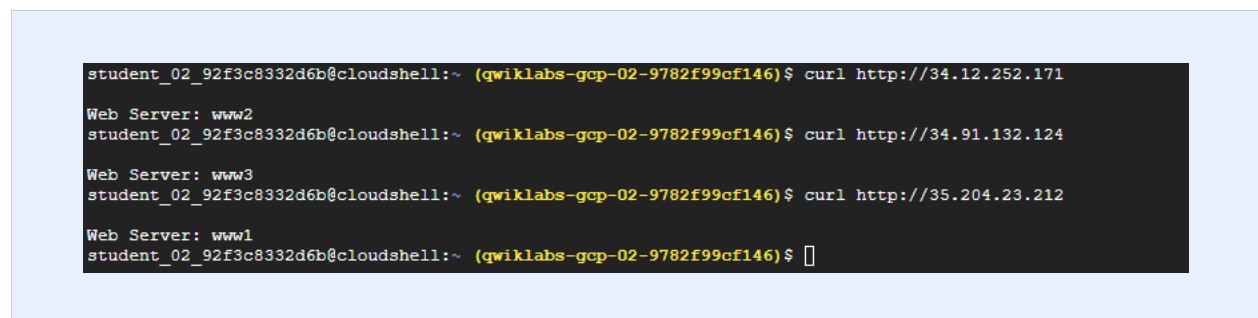
```
gcloud compute instances create www1 --zone=[ZONE] --tags=network-lb-tag --machine-type=e2-small --image-family=debian-11 --image-project=debian-cloud
gcloud compute instances create www2 --zone=[ZONE] --tags=network-lb-tag --machine-type=e2-small --image-family=debian-11 --image-project=debian-cloud
gcloud compute instances create www3 --zone=[ZONE] --tags=network-lb-tag --machine-type=e2-small --image-family=debian-11 --image-project=debian-cloud
gcloud compute firewall-rules create www-firewall-network-lb --target-tags network-lb-tag --allow tcp:80
```



```
CLOUD SHELL
Terminal (qwiklabs-gcp-04-2c17cc2e5e1e) x +
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more

Web Server: www1" | tee /var/www/html/index.html'
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-2c17cc2e5e1e/zones/europe-west4-a/instances/www1].
NAME: www1
ZONE: europe-west4-a
MACHINE TYPE: e2-small
PREEMPTIBLE:
INTERNAL IP: 10.164.0.4
EXTERNAL IP: 34.6.69.76
STATUS: RUNNING
student_02_8896023da0b@cloudshell:~ (qwiklabs-gcp-04-2c17cc2e5e1e)$ gcloud compute instances create www2 \
> --zone=europe-west4-a \
> --tags=network-lb-tag \
> --machine-type=e2-small \
> image-family=debian-11 \
>
ERROR: (gcloud.compute.instances.create) unrecognized arguments: image-family=debian-11 (did you mean '--image-family'?)

To search the help text of gcloud commands, run:
gcloud help -- SEARCH TERMS
student_02_8896023da0b@cloudshell:~ (qwiklabs-gcp-04-2c17cc2e5e1e)$ gcloud compute instances create ww2 \
> --zone=europe-west4-a \
> --tags=network-lb-tag \
> --machine-type=e2-small \
> --image-family=debian-11 \
> --image-project=debian-cloud \
> --metadata=startup-script='#!/bin/bash apt-get update apt-get install apache2 -y service apache2 restart echo " Web Server: www2" | tee /var/html/index.html '
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-2c17cc2e5e1e/zones/europe-west4-a/instances/ww2].
NAME: ww2
ZONE: europe-west4-a
MACHINE TYPE: e2-small
PREEMPTIBLE:
INTERNAL IP: 10.164.0.5
EXTERNAL IP: 34.12.237.215
STATUS: RUNNING
student_02_8896023da0b@cloudshell:~ (qwiklabs-gcp-04-2c17cc2e5e1e)$ gcloud compute instances create www3 \
> --zone=europe-west4-a \
> --tags=network-lb-tag \
> --machine-type=e2-small \
> --image-family=debian-11 \
> --image-project=debian-cloud \
> --metadata=startup-script='#!/bin/bash apt-get update install apache2 -y service apache2 restart echo " Web Server: www3" | tee /var/www/html/index.html'
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-04-2c17cc2e5e1e/zones/europe-west4-a/instances/www3].
NAME: www3
ZONE: europe-west4-a
MACHINE TYPE: e2-small
PREEMPTIBLE:
INTERNAL IP: 10.164.0.6
EXTERNAL IP: 34.6.19.138
STATUS: RUNNING
student_02_8896023da0b@cloudshell:~ (qwiklabs-gcp-04-2c17cc2e5e1e)$
```



```
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ curl http://34.12.252.171

Web Server: www2
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ curl http://34.91.132.124

Web Server: www3
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ curl http://35.204.23.212

Web Server: www1
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$
```

Task 3 — Configure the Load Balancing Service

Created a static external IP address for the load balancer and added a legacy HTTP health check to monitor instance health.

```
gcloud compute addresses create network-lb-ip-1 --region [REGION]
gcloud compute http-health-checks create basic-check
```

```
Web Server: www1
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ gcloud compute addresses create network-lb-ip-1 \
> --region europe-west4
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-02-9782f99cf146/regions/europe-west4/addresses/network-lb-ip-1].
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ gcloud compute http-health-checks create basic-check
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-02-9782f99cf146/global/httpHealthChecks/basic-check].
NAME: basic-check
HOST:
PORT: 80
REQUEST_PATH: /
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$
```

Task 4 — Create Target Pool & Forwarding Rule

A target pool groups the 3 backend VMs. The forwarding rule routes incoming traffic from the static IP to the target pool on port 80.

```
gcloud compute target-pools create www-pool --region [REGION] --http-health-check
basic-check
gcloud compute target-pools add-instances www-pool --instances www1,www2,www3
gcloud compute forwarding-rules create www-rule --region [REGION] --ports 80 --
address network-lb-ip-1 --target-pool www-pool
```

```
NAME: basic-check
HOST:
PORT: 80
REQUEST_PATH: /
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ gcloud compute target-pools create www-pool \
--region europe-west4 --http-health-check basic-check
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-02-9782f99cf146/regions/europe-west4/targetPools/www-pool].
NAME: www-pool
REGION: europe-west4
SESSION_AFFINITY: NONE
BACKUP:
HEALTH_CHECKS: basic-check
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ gcloud compute target-pools add-instances www-pool \
--instances www1,www2,www3
ERROR: (gcloud.compute.target-pools.add-instances) Could not fetch resource:
- The resource 'projects/qwiklabs-gcp-02-9782f99cf146/regions/europe-west4/targetPools/www-pool' was not found

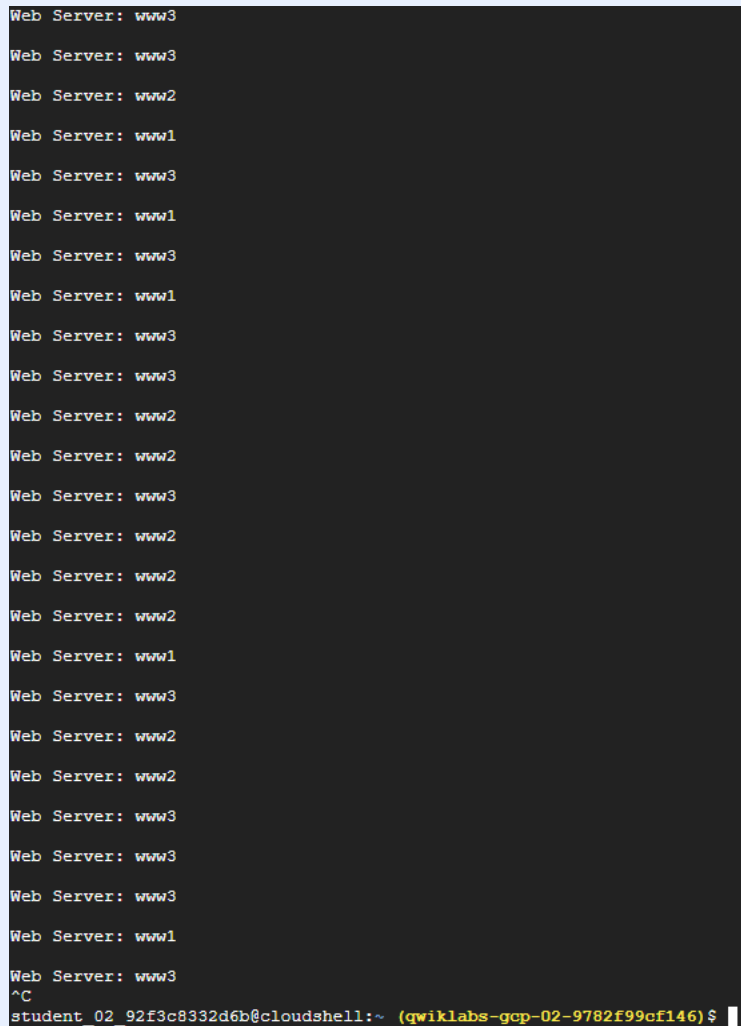
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ gcloud compute target-pools add-instances www-pool \
--instances www1, www2,www3
ERROR: (gcloud.compute.target-pools.add-instances) unrecognized arguments: www2,www3

To search the help text of gcloud commands, run:
gcloud help -- SEARCH TERMS
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ gcloud compute target-pools add-instances www-pool \
--instances www1,www2,www3
Updated [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-02-9782f99cf146/regions/europe-west4/targetPools/www-pool].
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$ gcloud compute forwarding-rules create www-rule \
--region europe-west4 \
--ports 80 \
--address network-lb-ip-1 \
--target-pool www-pool
Created [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-02-9782f99cf146/regions/europe-west4/forwardingRules/www-rule].
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$
```

Task 5 — Send Traffic to Instances

Used curl in a loop to send repeated requests to the load balancer external IP. The responses rotated between www1, www2, and www3 confirming traffic distribution is working.

```
IPADDRESS=$(gcloud compute forwarding-rules describe www-rule --region [REGION] --  
format="json" | jq -r .IPAddress)  
while true; do curl -m1 $IPADDRESS; done
```

A terminal window with a black background and yellow text. It shows a loop of curl requests to a load balancer's external IP. The responses, which are the 'Web Server' field from the JSON output, rotate between 'www1', 'www2', and 'www3'. The sequence of responses shown is: www3, www3, www2, www1, www3, www1, www3, www1, www3, www3, www2, www2, www3, www2, www2, www1, www3, www2, www2, www3, www3, www3, www1, www3. The terminal ends with a Ctrl-C (^C) and a shell prompt for a user named 'student_02_92f3c8332d6b' on a cloudshell.

```
Web Server: www3  
Web Server: www3  
Web Server: www2  
Web Server: www1  
Web Server: www3  
Web Server: www1  
Web Server: www3  
Web Server: www1  
Web Server: www3  
Web Server: www3  
Web Server: www2  
Web Server: www2  
Web Server: www3  
Web Server: www2  
Web Server: www2  
Web Server: www1  
Web Server: www3  
Web Server: www2  
Web Server: www2  
Web Server: www3  
Web Server: www3  
Web Server: www3  
Web Server: www1  
Web Server: www3  
^C  
student_02_92f3c8332d6b@cloudshell:~ (qwiklabs-gcp-02-9782f99cf146)$
```

What I Learned

- A Layer 4 NLB routes traffic based on IP/port — it does not inspect content
- Target pools group backend instances that receive incoming traffic

- Forwarding rules direct traffic from a static IP to a target pool
- Health checks automatically remove unhealthy instances from rotation
- All backend instances in a target pool must be in the same region

Challenges & How I Solved Them

- **Typos in curl command (Task 2)** — Mistyped the IP address (confused 1 and l, w errors) which caused curl to fail when verifying VMs were responding. Fixed by carefully re-reading the external IP from gcloud compute instances list and retyping it character by character.
- **Target pool issues (Task 4)** — Same typo problem carried over when setting up the forwarding rule and target pool. Learned to copy/paste IP addresses and resource names directly from the terminal output instead of retyping them to avoid errors.
- **Key Lesson** — In GCP commands, one wrong character breaks everything. Always copy/paste resource names and IP addresses rather than typing them manually.

Resources

GCP Network Load Balancer Docs: <https://cloud.google.com/load-balancing/docs/network>
Lab ID: GSP007